Universidade do Algarve

Faculdade de Ciências e Tecnologia

Licenciatura em Engenharia Informática

Redes de Computadores I

Laboratório 3

Wireshark - DNS

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PL: 2

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Relatório

1. nslookup

1. **Run nslookup to obtain the IP address of the web server for the Indian Institute of Technology in Bombay, India: www.iitb.ac.in. What is the IP address of www.iitb.ac.in**

R: O IP address de www.iitb.ac.in é 103.21.124.133.

2. **What is the IP address of the DNS server that provided the answer to your nslookup command in question 1 above?**

R: O IP address que providenciou a minha resposta à pergunta anterior foi 10.255.255.254 e a Porta é 53.

3. **Did the answer to your nslookup command in question 1 above come from an authoritative or non-authoritative server?**

R: As respostas vieram de um Non-authoritative server.

4. **Use the nslookup command to determine the name of the authoritative name server for the iit.ac.in domain. What is that name? (If there are more than one authoritative servers, what is the name of the first authoritative server returned by nslookup)? If you had to find the IP address of that authoritative name server, how would you do so?**

R: Apareceram 3 servidores no dominio iit.ac.in o nome do primeiro é dns3.iitb.ac.in., utilizava o comando "nslookup dns1.iitb.ac.in" para encontrar o IP adress que neste caso é 103.21.125.129.

```
carva@Carvalho:/mnt/c/Users/carva$ nslookup www.iitb.ac.in
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   www.iitb.ac.in
Address: 103.21.124.133
;; Got recursion not available from 10.255.255.254

carva@Carvalho:/mnt/c/Users/carva$ nslookup -type=NS iitb.ac.in
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
iitb.ac.in      nameserver = dns1.iitb.ac.in.
iitb.ac.in      nameserver = dns2.iitb.ac.in.
iitb.ac.in      nameserver = dns3.iitb.ac.in.

Authoritative answers can be found from:
dns1.iitb.ac.in internet address = 103.21.125.129
dns2.iitb.ac.in internet address = 103.21.126.129
dns3.iitb.ac.in internet address = 103.21.127.129

carva@Carvalho:/mnt/c/Users/carva$ nslookup dns3.iitb.ac.in
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   dns3.iitb.ac.in
Address: 103.21.127.129
```

2. The DNS cache on your computer

Para limpar a DNS cache basta utilizar:

No Mac – “sudo killall -HUP mDNSResponder”

No Windows – “ipconfig /flushdns”

No Linux – “sudo systemd-resolve --flush-caches”

No Ubuntu(v22.04 and later) – “sudo resolvectl flush-caches”

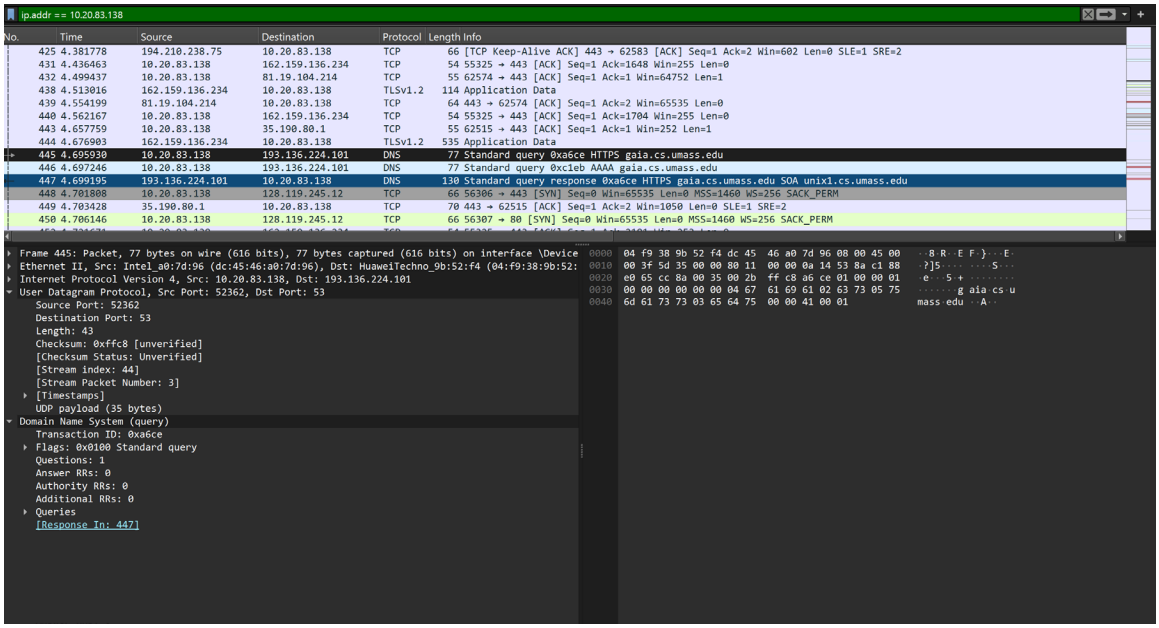
```
carva@Carvalho:/mnt/c/Users/carva$ sudo resolvectl flush-caches
```

3. Tracing DNS with Wireshark

(ip.addr == 10.20.83.138)

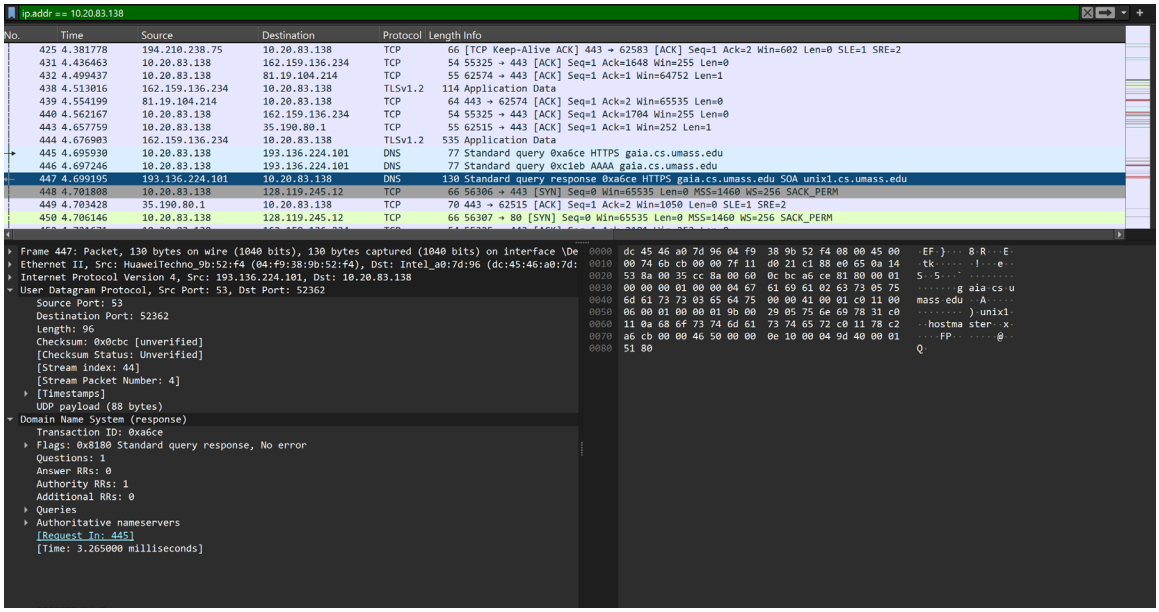
5. **Locate the first DNS query message resolving the name gaia.cs.umass.edu. What is the packet number in the trace for the DNS query message? Is this query message sent over UDP or TCP?**

R: É o pacote numero 445, e foi enviado por UDP.



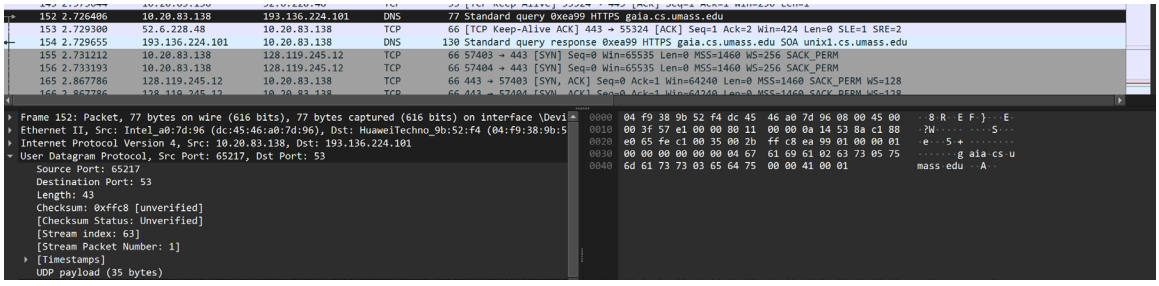
6. **Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?**

R: O numero do pacote foi o 447, a mensagem foi recebida por UDP.



7. **What is the destination port for the DNS query message? What is the source port of the DNS response message?**

R: A porta destino é 53, e a porta origem é 53.

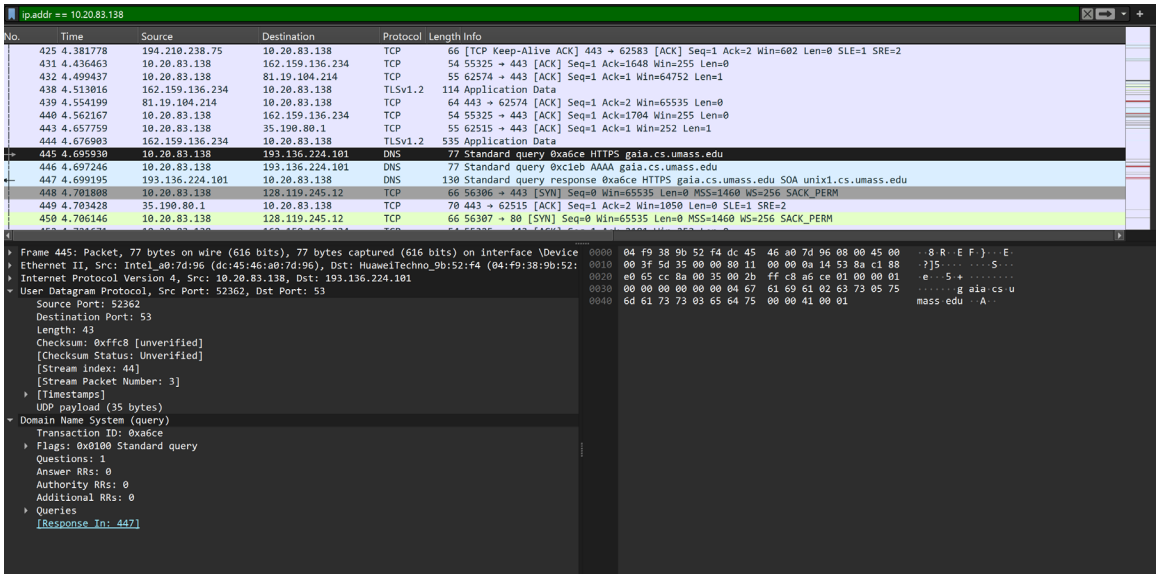


8. **To what IP address is the DNS query message sent?**

R: A query foi enviada para o IP address 193.136.224.101.

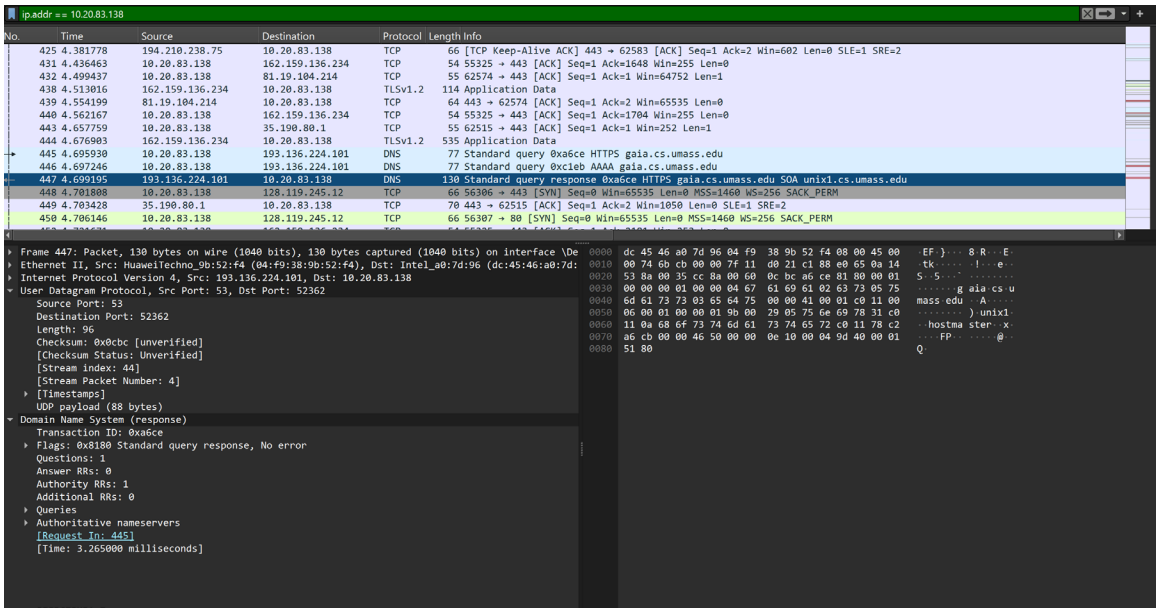
9. **Examine the DNS query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?**

R: A DNS query message, contem 1 pergunta e 0 respostas.



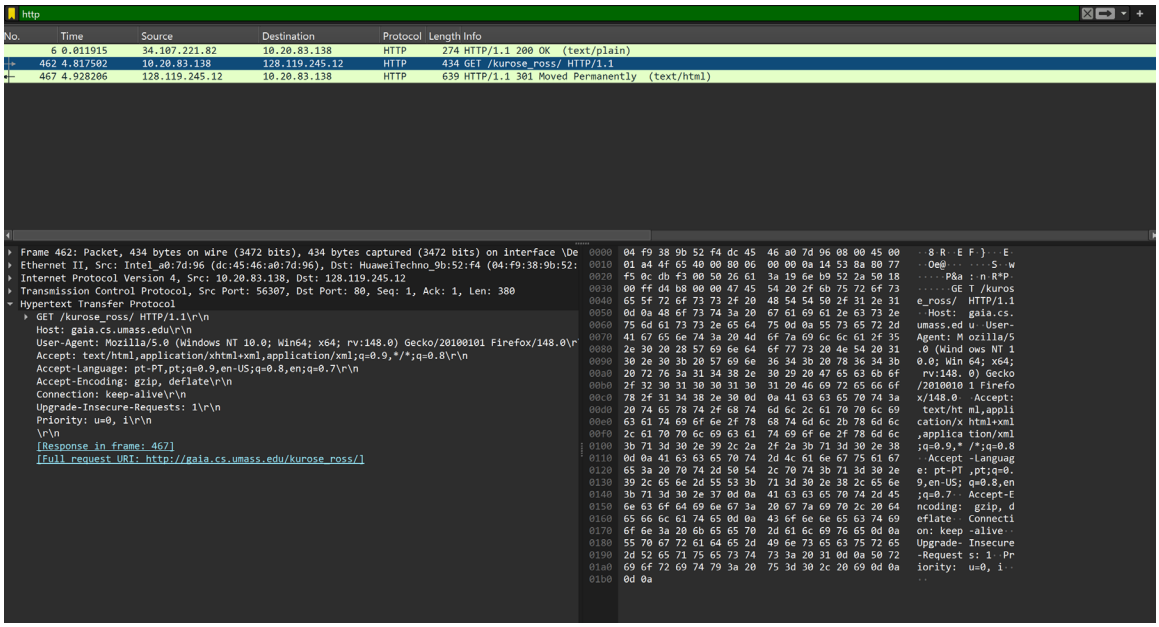
10. **Examine the DNS response message to the initial query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?**

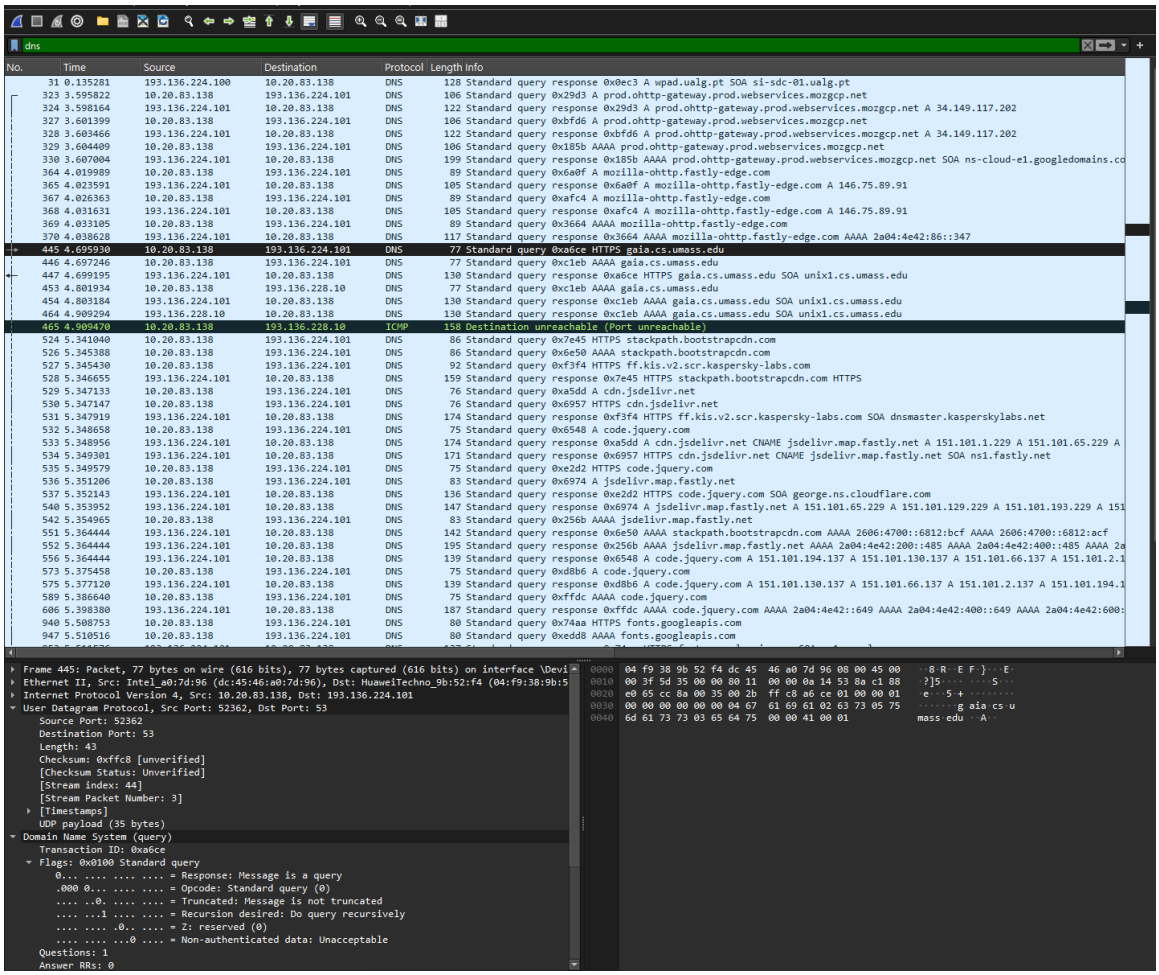
R: A mensagem de resposta do DNS à query inicial, contem 1 pergunta e 0 respostas.



11. **The web page for the base file http://gaia.cs.umass.edu/kurose_ross/ references the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E_2.jpg , which, like the base webpage, is on gaia.cs.umass.edu. What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/? What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address? What is the packet number in the trace of the received DNS response? What is the packet number in the trace for the HTTP GET request for the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E2.jpg? What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address? Discuss how DNS caching affects the answer to this last question.**

R: O pacote do HTTP GET é 462, O pacote do DNS inicial é 445 e o pacote da resposta é 447, O pacote com o HTTP GET da imagem não aparece no trace porque o servidor redireciona o acesso para HTTPS. A partir daí, o pedido da imagem é feito sobre TLS e fica cifrado, logo o Wireshark não mostra o “HTTP GET …jpg” em claro.

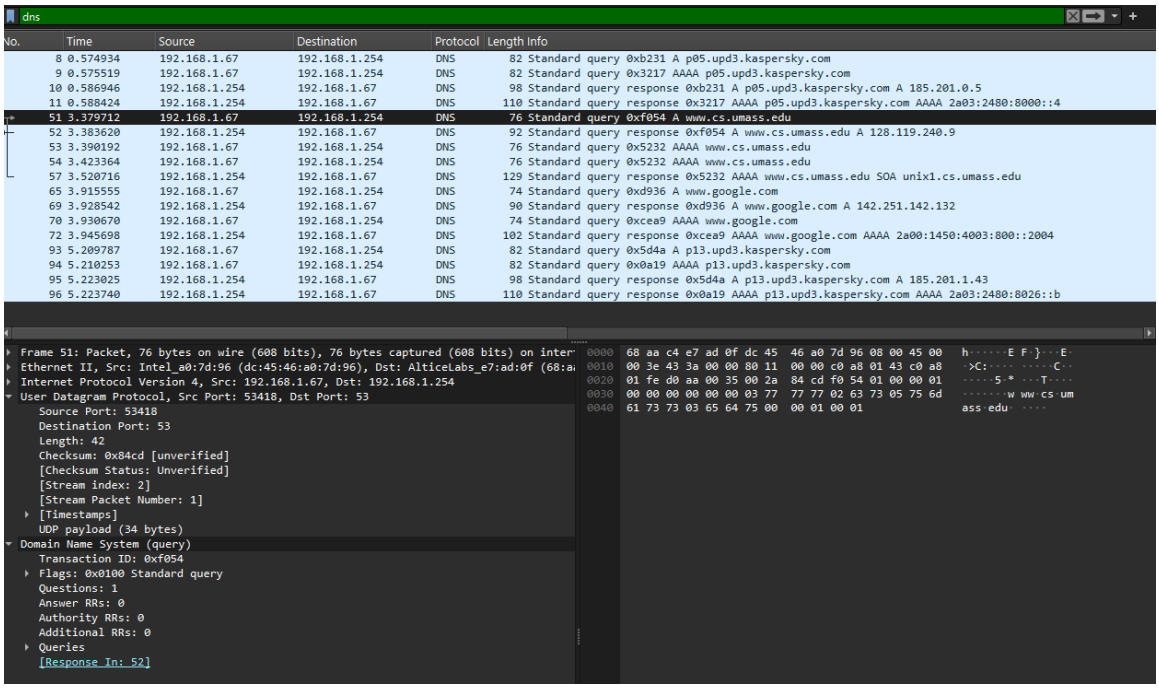

http
No. Time Source Destination Protocol Length Info
6 0.011915 34.107.221.82 10.20.83.138 HTTP 274 HTTP/1.1 200 OK (text/plain)
462 4.817502 10.20.83.138 128.119.245.12 HTTP 434 GET /kurose_ross/ HTTP/1.1
467 4.928206 128.119.245.12 10.20.83.138 HTTP 639 HTTP/1.1 301 Moved Permanently (text/html)
Frame 462: Packet, 434 bytes on wire (3472 bits), 434 bytes captured (3472 bits) on interface \De
Ethernet II, Src: Intel_a0:7d:96 (dc:45:46:a0:7d:96), Dst: HuaweiTechno_9b:52:f4 (04:f9:38:9b:52:
Internet Protocol Version 4, Src: 10.20.83.138, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 56307, Dst Port: 80, Seq: 1, Ack: 1, Len: 380
Hypertext Transfer Protocol
GET /kurose_ross/ HTTP/1.1\r\n
Host: gaia.cs.umass.edu\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:148.0) Gecko/20100101 Firefox/148.0\r
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
Accept-Language: pt-PT,pt;q=0.9,en-US;q=0.8,en;q=0.7\r\n
Accept-Encoding: gzip, deflate\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
Priority: u=0, i\r\n
\r\n
[Response in frame: 467]
[Full request URI: http://gaia.cs.umass.edu/kurose_ross/]

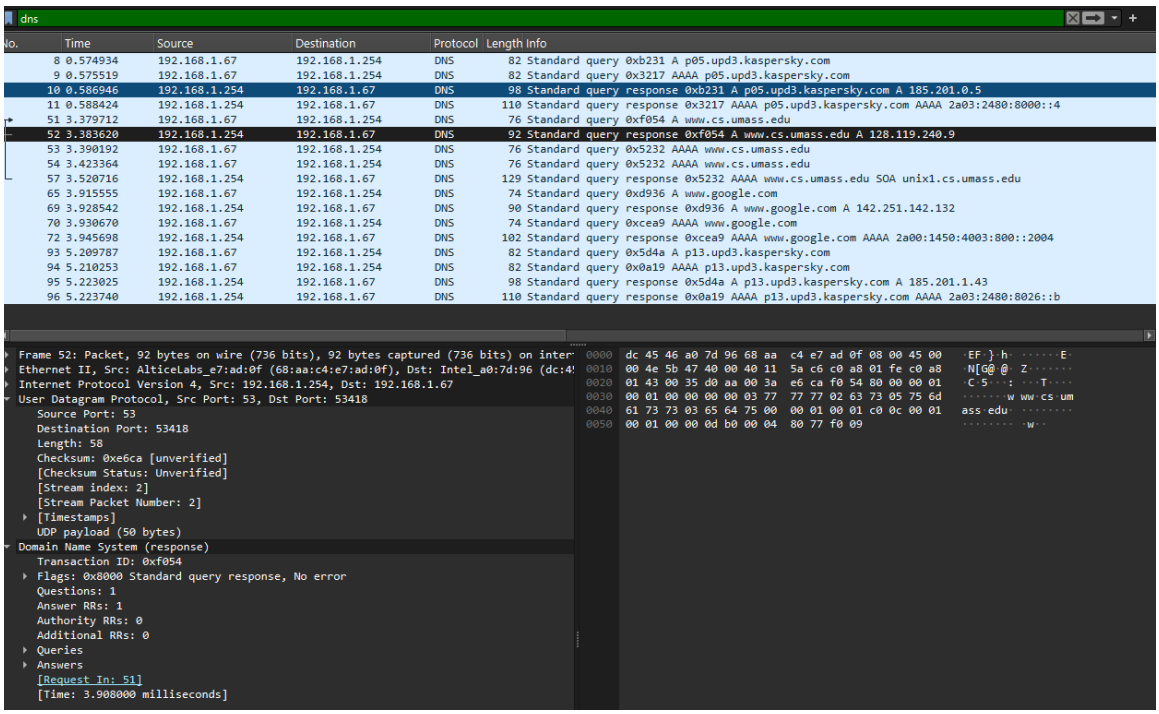

dns
No. Time Source Destination Protocol Length Info
31 0.135281 193.136.224.100 10.20.83.138 DNS 128 Standard query response 0x0ec3 A wpad.ualg.pt SOA si-sdc-01.ualg.pt
323 3.595822 10.20.83.138 193.136.224.101 DNS 106 Standard query 0x29d3 A prod.ohttp-gateway.prod.webservices.mozgcp.net
324 3.598164 193.136.224.101 10.20.83.138 DNS 122 Standard query response 0x29d3 A prod.ohttp-gateway.prod.webservices.mozgcp.net A 34.149.117.202
327 3.601399 10.20.83.138 193.136.224.101 DNS 106 Standard query 0xbfd6 A prod.ohttp-gateway.prod.webservices.mozgcp.net
328 3.603466 193.136.224.101 10.20.83.138 DNS 122 Standard query response 0xbfd6 A prod.ohttp-gateway.prod.webservices.mozgcp.net A 34.149.117.202
329 3.604409 10.20.83.138 193.136.224.101 DNS 106 Standard query 0x185b AAAA prod.ohttp-gateway.prod.webservices.mozgcp.net
330 3.607004 193.136.224.101 10.20.83.138 DNS 199 Standard query response 0x185b AAAA prod.ohttp-gateway.prod.webservices.mozgcp.net SOA ns-cloud-e1.googledomains.co
364 4.019989 10.20.83.138 193.136.224.101 DNS 89 Standard query 0x6a0f A mozilla-ohttp.fastly-edge.com
365 4.023591 193.136.224.101 10.20.83.138 DNS 105 Standard query response 0x6a0f A mozilla-ohttp.fastly-edge.com A 146.75.89.91
367 4.026363 10.20.83.138 193.136.224.101 DNS 89 Standard query 0xafc4 A mozilla-ohttp.fastly-edge.com
368 4.031631 193.136.224.101 10.20.83.138 DNS 105 Standard query response 0xafc4 A mozilla-ohttp.fastly-edge.com A 146.75.89.91
369 4.033105 10.20.83.138 193.136.224.101 DNS 89 Standard query 0x3664 AAAA mozilla-ohttp.fastly-edge.com
370 4.038628 193.136.224.101 10.20.83.138 DNS 117 Standard query response 0x3664 AAAA mozilla-ohttp.fastly-edge.com AAAA 2a04:4e42:86::347
445 4.695930 10.20.83.138 193.136.224.101 DNS 77 Standard query 0xa6ce HTTPS gaia.cs.umass.edu
446 4.697246 10.20.83.138 193.136.224.101 DNS 77 Standard query 0xc1eb AAAA gaia.cs.umass.edu
447 4.699195 193.136.224.101 10.20.83.138 DNS 130 Standard query response 0xa6ce HTTPS gaia.cs.umass.edu SOA unix1.cs.umass.edu
453 4.801934 10.20.83.138 193.136.228.10 DNS 77 Standard query 0xc1eb AAAA gaia.cs.umass.edu
454 4.803184 193.136.224.101 10.20.83.138 DNS 130 Standard query response 0xc1eb AAAA gaia.cs.umass.edu SOA unix1.cs.umass.edu
464 4.909294 193.136.228.10 10.20.83.138 DNS 130 Standard query response 0xc1eb AAAA gaia.cs.umass.edu SOA unix1.cs.umass.edu
465 4.909470 10.20.83.138 193.136.228.10 ICMP 158 Destination unreachable (Port unreachable)
524 5.341040 10.20.83.138 193.136.224.101 DNS 86 Standard query 0x7e45 HTTPS stackpath.bootstrapcdn.com
526 5.345388 10.20.83.138 193.136.224.101 DNS 86 Standard query 0x6e50 AAAA stackpath.bootstrapcdn.com
527 5.345430 10.20.83.138 193.136.224.101 DNS 92 Standard query 0xf3f4 HTTPS ff.kis.v2.scr.kaspersky-labs.com
528 5.346655 193.136.224.101 10.20.83.138 DNS 159 Standard query response 0x7e45 HTTPS stackpath.bootstrapcdn.com HTTPS
529 5.347133 10.20.83.138 193.136.224.101 DNS 76 Standard query 0xa5dd A cdn.jsdelivr.net
530 5.347147 10.20.83.138 193.136.224.101 DNS 76 Standard query 0x6957 HTTPS cdn.jsdelivr.net
531 5.347919 193.136.224.101 10.20.83.138 DNS 174 Standard query response 0xf3f4 HTTPS ff.kis.v2.scr.kaspersky-labs.com SOA dnsmaster.kasperskylabs.net
532 5.348658 10.20.83.138 193.136.224.101 DNS 75 Standard query 0x6548 A code.jquery.com
533 5.348956 193.136.224.101 10.20.83.138 DNS 174 Standard query response 0xa5dd A cdn.jsdelivr.net CNAME jsdelivr.map.fastly.net A 151.101.1.229 A 151.101.65.229 A
534 5.349301 193.136.224.101 10.20.83.138 DNS 171 Standard query response 0x6957 HTTPS cdn.jsdelivr.net CNAME jsdelivr.map.fastly.net SOA ns1.fastly.net
535 5.349579 10.20.83.138 193.136.224.101 DNS 75 Standard query 0xe2d2 HTTPS code.jquery.com
536 5.351206 10.20.83.138 193.136.224.101 DNS 83 Standard query 0x6974 A jsdelivr.map.fastly.net
537 5.352143 193.136.224.101 10.20.83.138 DNS 136 Standard query response 0xe2d2 HTTPS code.jquery.com SOA george.ns.cloudflare.com
540 5.353952 193.136.224.101 10.20.83.138 DNS 147 Standard query response 0x6974 A jsdelivr.map.fastly.net A 151.101.65.229 A 151.101.129.229 A 151.101.193.229 A 151
542 5.354965 10.20.83.138 193.136.224.101 DNS 83 Standard query 0x256b AAAA jsdelivr.map.fastly.net
551 5.364444 193.136.224.101 10.20.83.138 DNS 142 Standard query response 0x6e50 AAAA stackpath.bootstrapcdn.com AAAA 2606:4700::6812:bcf AAAA 2606:4700::6812:acf
552 5.364444 193.136.224.101 10.20.83.138 DNS 195 Standard query response 0x256b AAAA jsdelivr.map.fastly.net AAAA 2a04:4e42:200::485 AAAA 2a04:4e42:400::485 AAAA 2a
556 5.364444 193.136.224.101 10.20.83.138 DNS 139 Standard query response 0x6548 A code.jquery.com A 151.101.194.137 A 151.101.130.137 A 151.101.66.137 A 151.101.2.1
573 5.375458 10.20.83.138 193.136.224.101 DNS 75 Standard query 0xd8b6 A code.jquery.com
575 5.377120 193.136.224.101 10.20.83.138 DNS 139 Standard query response 0xd8b6 A code.jquery.com A 151.101.130.137 A 151.101.66.137 A 151.101.2.137 A 151.101.194.1
589 5.386640 10.20.83.138 193.136.224.101 DNS 75 Standard query 0xffdc AAAA code.jquery.com
606 5.398380 193.136.224.101 10.20.83.138 DNS 187 Standard query response 0xffdc AAAA code.jquery.com AAAA 2a04:4e42::649 AAAA 2a04:4e42:400::649 AAAA 2a04:4e42:600:
940 5.508753 10.20.83.138 193.136.224.101 DNS 80 Standard query 0x74aa HTTPS fonts.googleapis.com
947 5.510516 10.20.83.138 193.136.224.101 DNS 80 Standard query 0xedd8 AAAA fonts.googleapis.com
Frame 445: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \Devi
Ethernet II, Src: Intel_a0:7d:96 (dc:45:46:a0:7d:96), Dst: HuaweiTechno_9b:52:f4 (04:f9:38:9b:5
Internet Protocol Version 4, Src: 10.20.83.138, Dst: 193.136.224.101
User Datagram Protocol, Src Port: 52362, Dst Port: 53
Source Port: 52362
Destination Port: 53
Length: 43
Checksum: 0xffc8 [unverified]
[Checksum Status: Unverified]
[Stream index: 44]
[Stream Packet Number: 3]
[Timestamps]
UDP payload (35 bytes)
Domain Name System (query)
Transaction ID: 0xa6ce
Flags: 0x0100 Standard query
0... .... = Response: Message is a query
.000 0... .... = Opcode: Standard query (0)
.... ..0. .... = Truncated: Message is not truncated
....1 .... = Recursion desired: Do query recursively
.... .0.. .... = Z: reserved (0)
....0 .... = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0

Executando – “nslookup www.cs.umass.edu”

12. What is the destination port for the DNS query message? What is the source port of the DNS response message?

R: A porta de destino da mensagem query DNS é 53, e a porta de origem da resposta a mensagem query DNS é 53.





13. **To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?**

R: A mensagem query DNS foi enviada para o IP address 193.136.224.101, sim este IP address é o default.

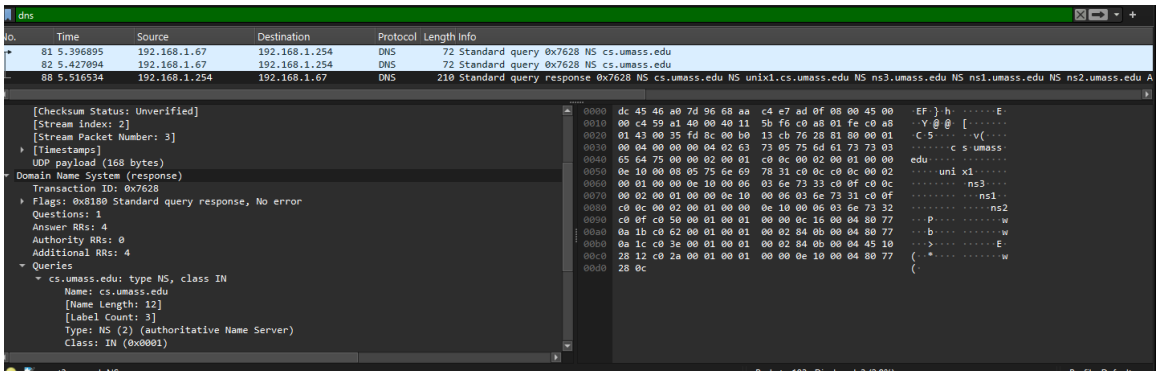
14. **Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?**

R: A query DNS é Type A, e não contem nenhuma pergunta.

Executando – "nslookup –type=NS umass.edu"

15. **Examine the DNS response message to the query message. How many "questions" does this DNS response message contain? How many "answers"?**

R: A DNS query message, contem 1 pergunta e 4 respostas.

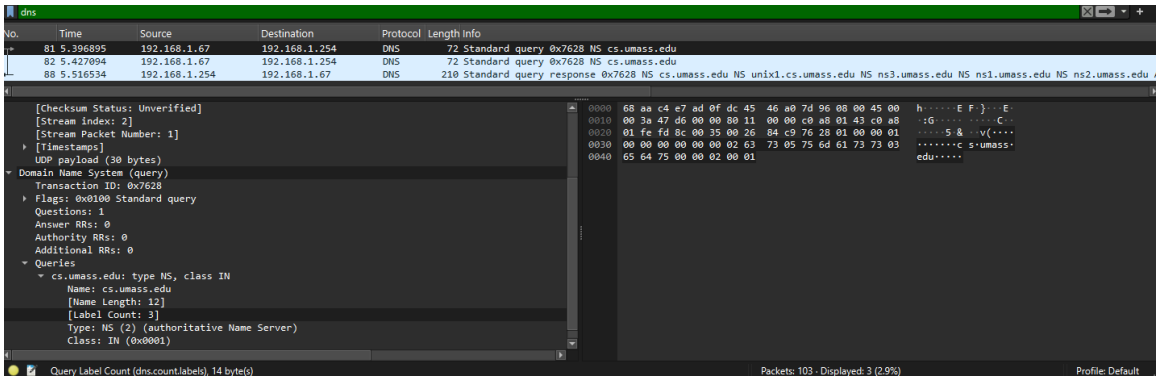


16. **To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?**

R: A mensagem query DNS foi enviada para o IP address 192.168.1.254, sim este IP address é o default.

17. **Examine the DNS query message. How many questions does the query have? Does the query message contain any "answers"?**

R: A DNS query message, contem 1 pergunta e 0 respostas.



18. **Examine the DNS response message (in particular the DNS response message that has type "NS"). How many answers does the response have? What information is contained in the answers? How many additional resource records are returned? What additional information is included in these additional resource records (if additional information is returned)?**

R: A DNS query message, contem 4 respostas,

A informaçao contida nas respostas é:

unix1.cs.umass.edu,

ns3.umass.edu,

ns1.umass.edu,

ns2.umass.edu;

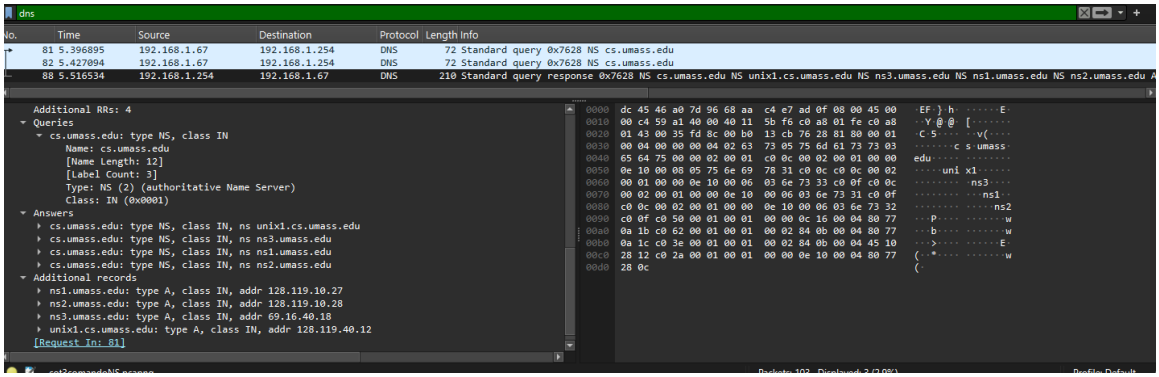
Foram returnados 4 registros de recursos adicionais, com a informaçao seguinte:

ns1.umass.edu → **128.119.10.27**

ns2.umass.edu → **128.119.10.28**

ns3.umass.edu → **69.16.40.18**

unix1.cs.umass.edu → **128.119.40.12**


dns
No. Time Source Destination Protocol Length Info
81 5.396895 192.168.1.67 192.168.1.254 DNS 72 Standard query 0x7628 NS cs.umass.edu
82 5.427094 192.168.1.67 192.168.1.254 DNS 72 Standard query 0x7628 NS cs.umass.edu
88 5.516534 192.168.1.254 192.168.1.67 DNS 210 Standard query response 0x7628 NS cs.umass.edu NS unix1.cs.umass.edu NS ns3.umass.edu NS ns1.umass.edu NS ns2.umass.edu A
Additional RRs: 4
Queries
cs.umass.edu: type NS, class IN
Name: cs.umass.edu
[Name Length: 12]
[Label Count: 3]
Type: NS (2) (authoritative Name Server)
Class: IN (0x0001)
Answers
cs.umass.edu: type NS, class IN, ns unix1.cs.umass.edu
cs.umass.edu: type NS, class IN, ns ns3.umass.edu
cs.umass.edu: type NS, class IN, ns ns1.umass.edu
cs.umass.edu: type NS, class IN, ns ns2.umass.edu
Additional records
ns1.umass.edu: type A, class IN, addr 128.119.10.27
ns2.umass.edu: type A, class IN, addr 128.119.10.28
ns3.umass.edu: type A, class IN, addr 69.16.40.18
unix1.cs.umass.edu: type A, class IN, addr 128.119.40.12
[Request In: 81]
0000 dc 45 46 a0 7d 96 68 aa c4 e7 ad 0f 08 00 45 00
0010 00 c4 59 a1 40 00 40 11 5b f6 c0 a8 01 fe c0 a8
0020 01 43 00 35 fd 8c 00 b0 13 cb 76 28 81 80 00 01
0030 00 04 00 00 00 04 02 63 73 05 75 6d 61 73 73 03
0040 65 64 75 00 00 02 00 01 c0 0c 00 02 00 01 00 00
0050 0e 10 00 08 05 75 6e 69 78 31 c0 0c c0 0c 00 02
0060 00 01 00 00 0e 10 00 06 03 6e 73 33 c0 0f c0 0c
0070 00 02 00 01 00 00 0e 10 00 06 03 6e 73 31 c0 0f
0080 c0 0c 00 02 00 01 00 00 0e 10 00 06 03 6e 73 32
0090 c0 0f c0 50 00 01 00 01 00 00 0c 16 00 04 80 77
00a0 0a 1b c0 62 00 01 00 01 00 02 84 0b 00 04 80 77
00b0 0a 1c c0 3e 00 01 00 01 00 02 84 0b 00 04 45 10
00c0 28 12 c0 2a 00 01 00 01 00 00 0e 10 00 04 80 77
00d0 28 0c